LAB PROGRAMS 1-10

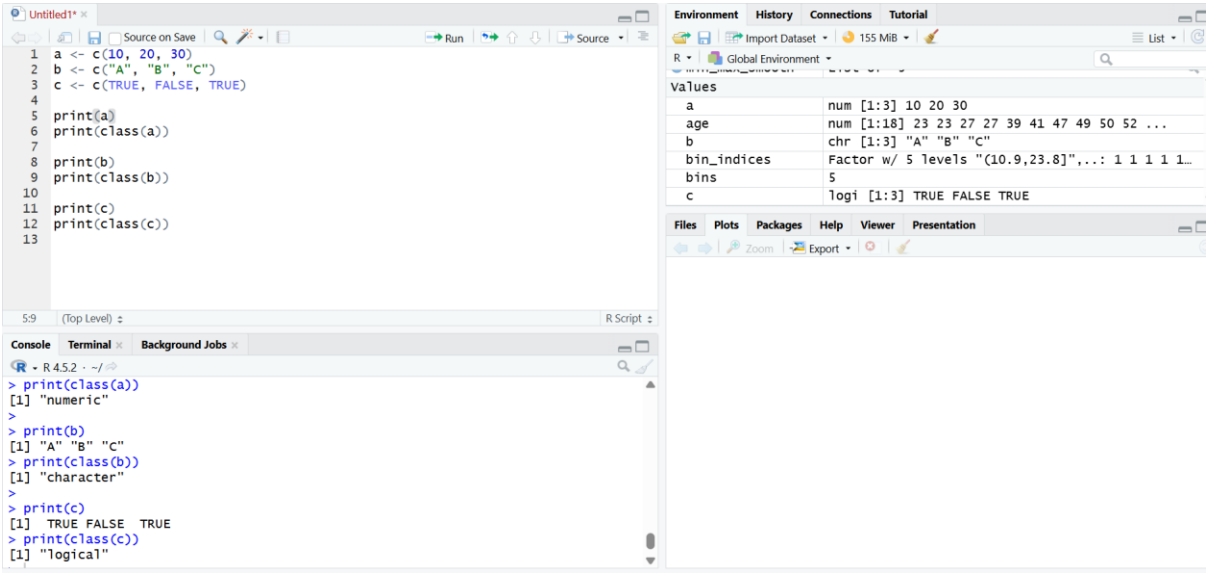
NAME: P.Sharmila

REG NO: 1921260281

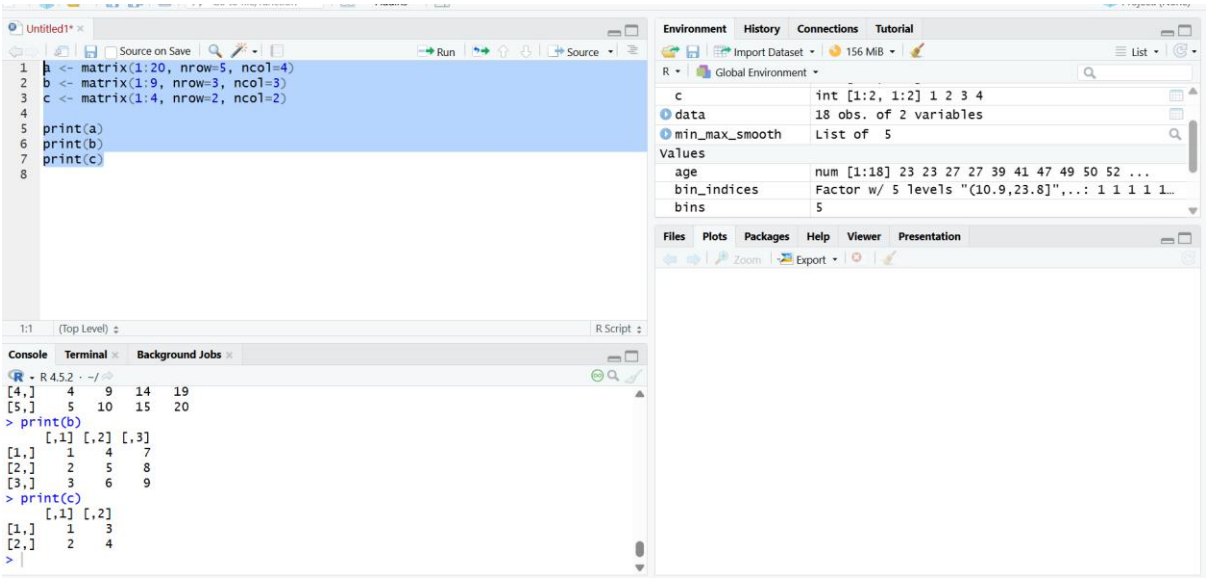
COURSE CODE:ITA0408

COURSE: STATISTICS WITH R

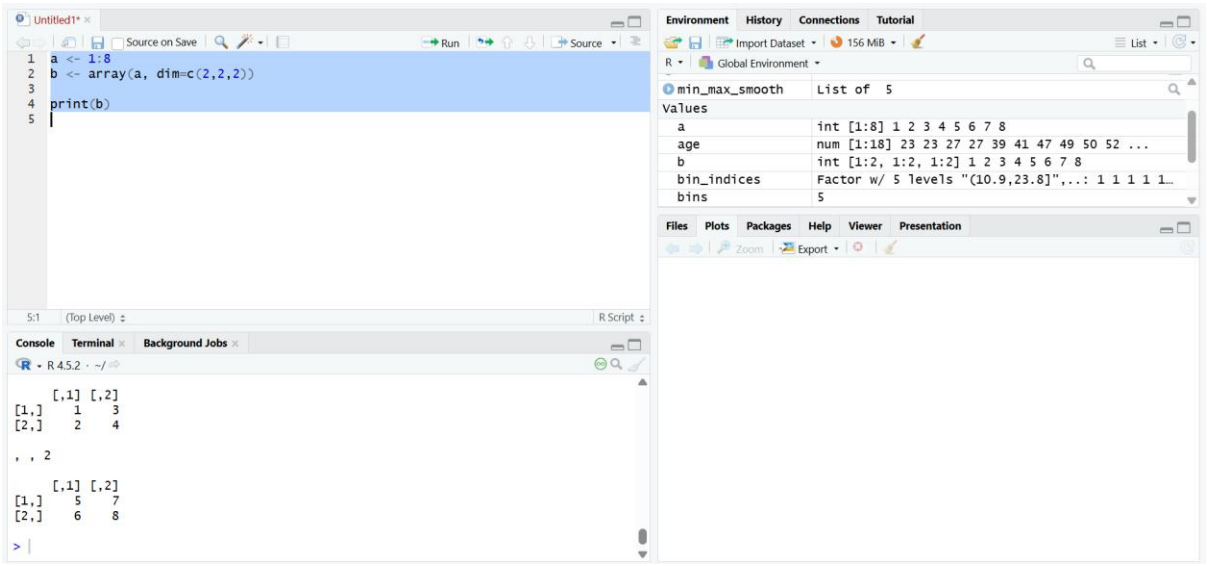
PROGRAMMING 1}



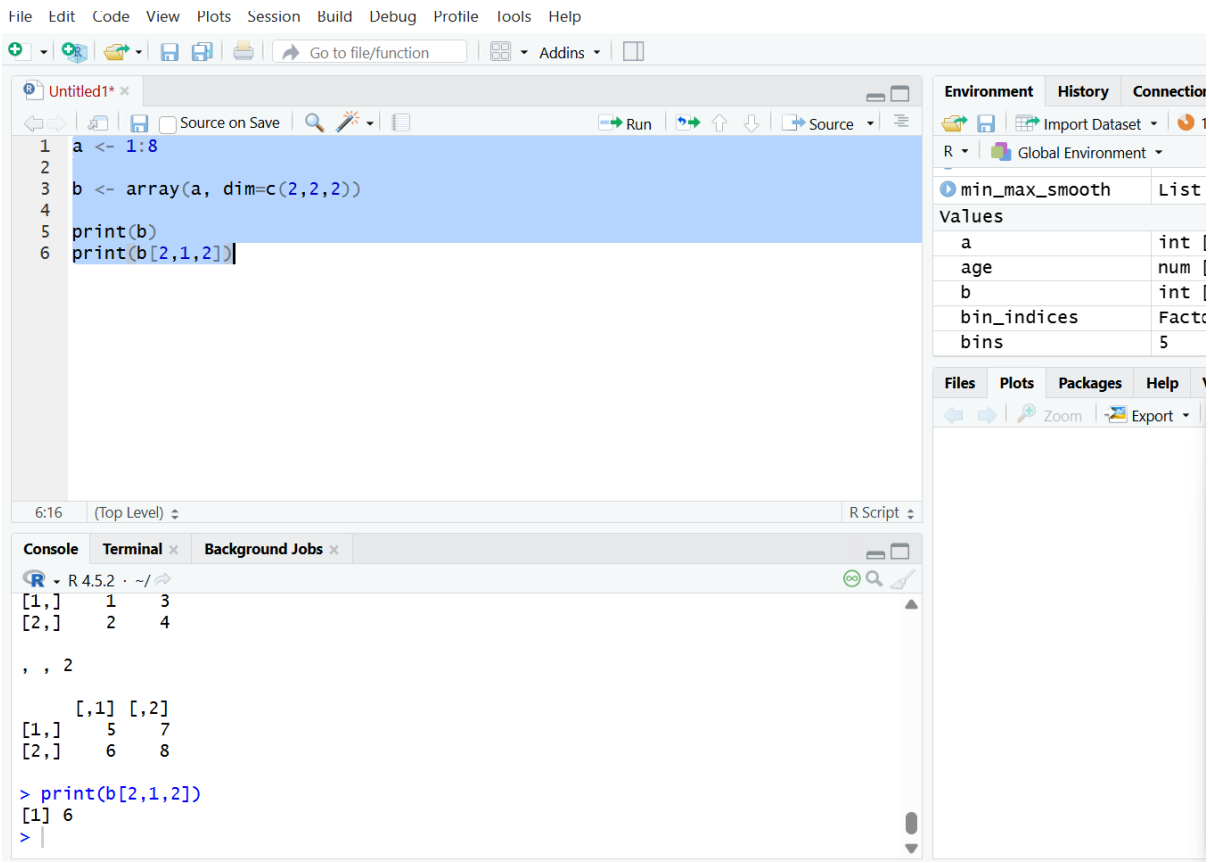
2}



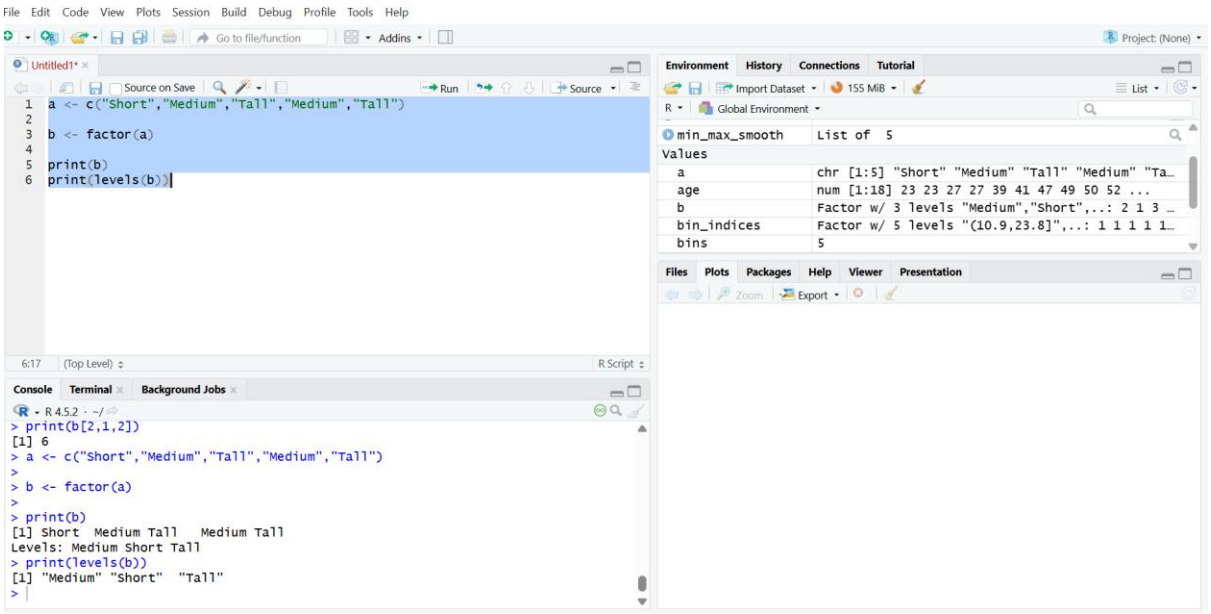
3}


Untitled1*
Source on Save
Run
Source
1 a <- 1:8
2 b <- array(a, dim=c(2,2,2))
3
4 print(b)
5
5:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.5.2 · ~/
[,1] [,2]
[1,] 1 3
[2,] 2 4
, , 2
[,1] [,2]
[1,] 5 7
[2,] 6 8
>
Environment
History
Connections
Tutorial
Import Dataset
156 MiB
List
R
Global Environment
min_max_smooth List of 5
Values
a int [1:8] 1 2 3 4 5 6 7 8
age num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
b int [1:2, 1:2, 1:2] 1 2 3 4 5 6 7 8
bin_indices Factor w/ 5 levels "(10.9,23.8]",..: 1 1 1 1 1...
bins 5
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export

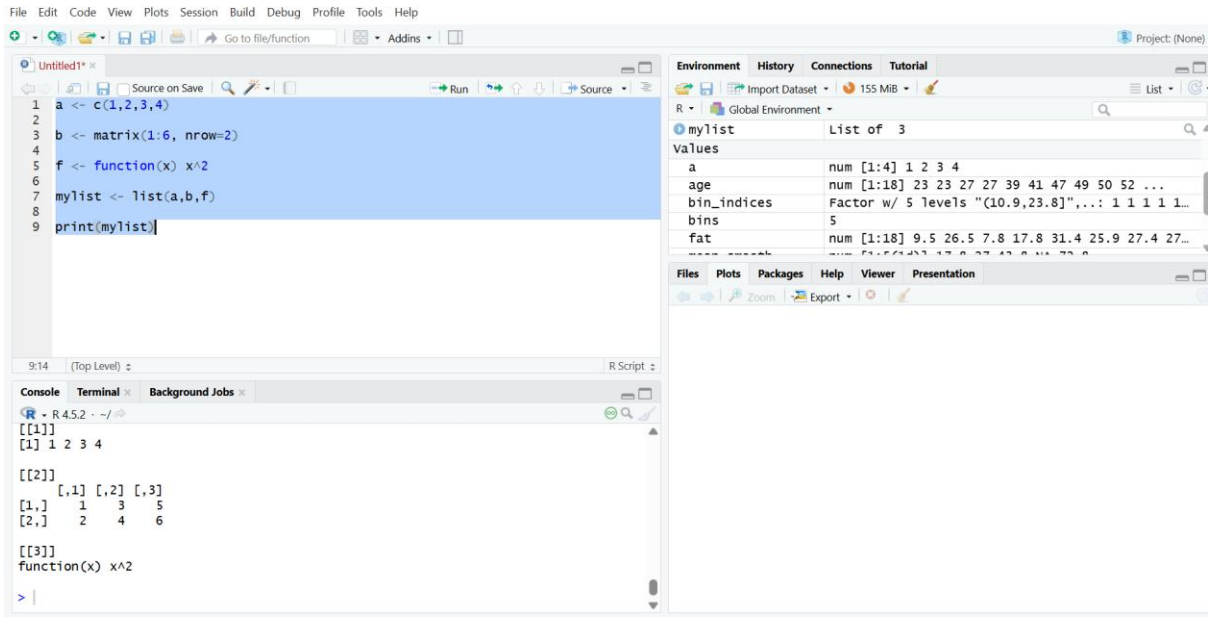
4}


File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled1*
Source on Save
Run
Source
1 a <- 1:8
2
3 b <- array(a, dim=c(2,2,2))
4
5 print(b)
6 print(b[2,1,2])
6:16
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.5.2 · ~/
[1,] 1 3
[2,] 2 4
, , 2
[,1] [,2]
[1,] 5 7
[2,] 6 8
> print(b[2,1,2])
[1] 6
>
Environment
History
Import Dataset
R
Global Environment
min_max_smooth
Values
a
age
b
bin_indices
bins
Files
Plots
Packages
Help
Zoom
Export

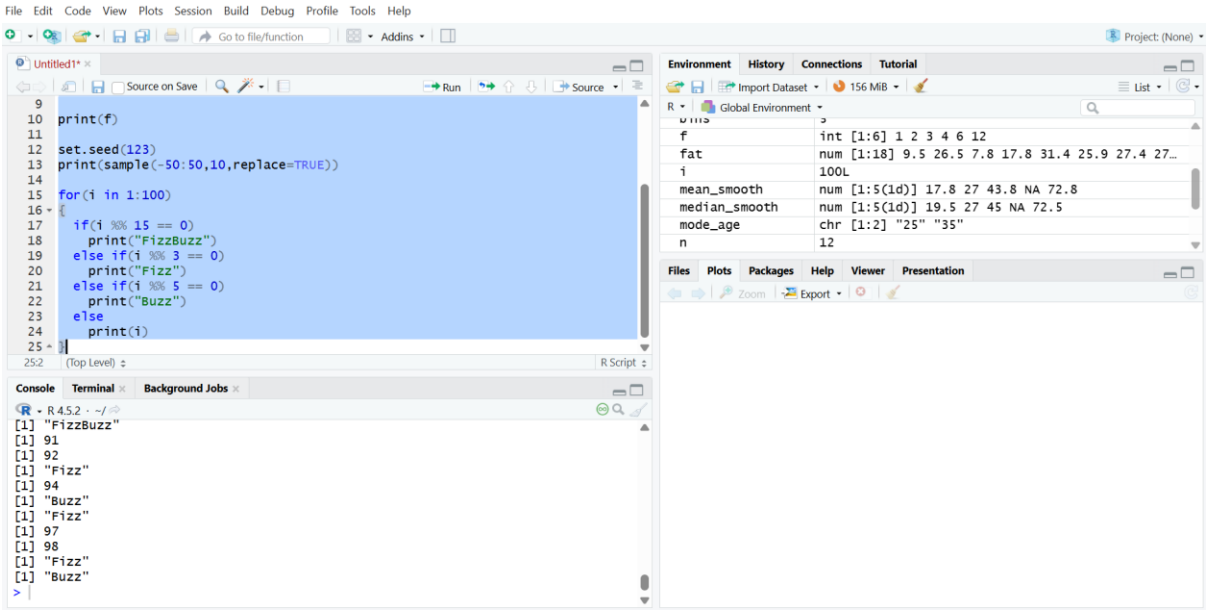
5}


File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 a <- c("Short","Medium","Tall","Medium","Tall")
2
3 b <- factor(a)
4
5 print(b)
6 print(levels(b))
6:17 (Top Level) R Script
Console Terminal Background Jobs
R 4.5.2 · ~/
> print(b[2,1,2])
[1] 6
> a <- c("Short","Medium","Tall","Medium","Tall")
>
> b <- factor(a)
>
> print(b)
[1] Short Medium Tall Medium Tall
Levels: Medium Short Tall
> print(levels(b))
[1] "Medium" "Short" "Tall"
>
Environment History Connections Tutorial
Import Dataset 155 MiB
R Global Environment
min_max_smooth List of 5
Values
a chr [1:5] "Short" "Medium" "Tall" "Medium" "Ta…
age num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
b Factor w/ 3 levels "Medium","Short",..: 2 1 3 …
bin_indices Factor w/ 5 levels "(10.9,23.8]",..: 1 1 1 1 1…
bins 5
Files Plots Packages Help Viewer Presentation
Zoom Export

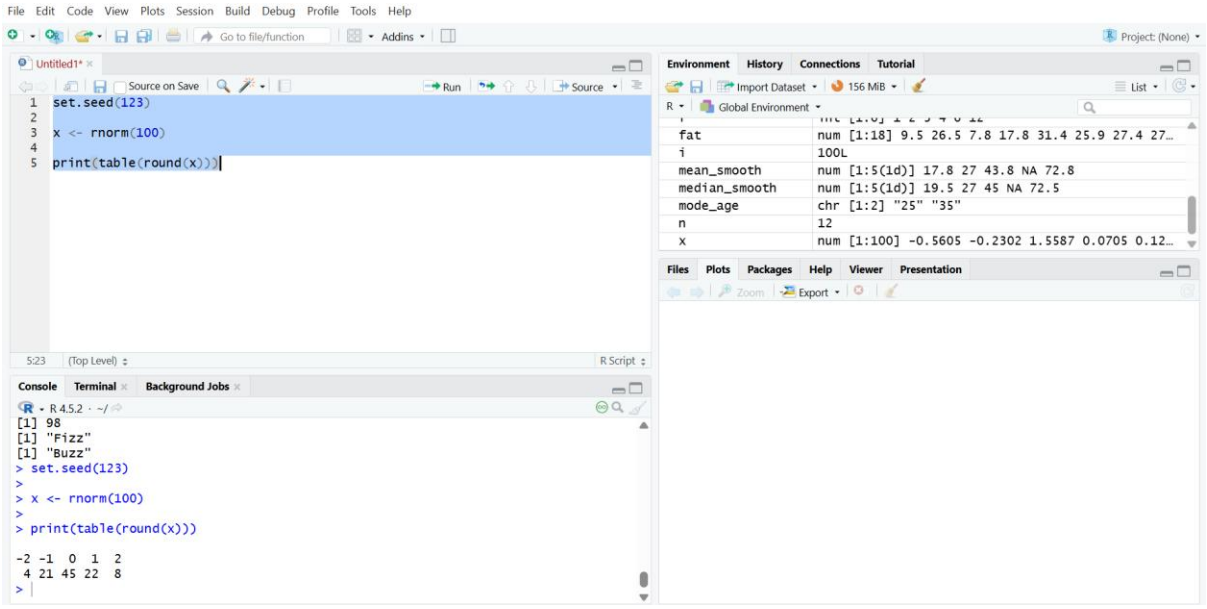
6}


File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 a <- c(1,2,3,4)
2
3 b <- matrix(1:6, nrow=2)
4
5 f <- function(x) x^2
6
7 mylist <- list(a,b,f)
8
9 print(mylist)
9:14 (Top Level) R Script
Console Terminal Background Jobs
R 4.5.2 · ~/
[[1]]
[1] 1 2 3 4
[[2]]
[,1] [,2] [,3]
[1,] 1 3 5
[2,] 2 4 6
[[3]]
function(x) x^2
>
Environment History Connections Tutorial
Import Dataset 155 MiB
R Global Environment
mylist List of 3
Values
a num [1:4] 1 2 3 4
age num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
bin_indices Factor w/ 5 levels "(10.9,23.8]",..: 1 1 1 1 1…
bins 5
fat num [1:18] 9.5 26.5 7.8 17.8 31.4 25.9 27.4 27…
Files Plots Packages Help Viewer Presentation
Zoom Export

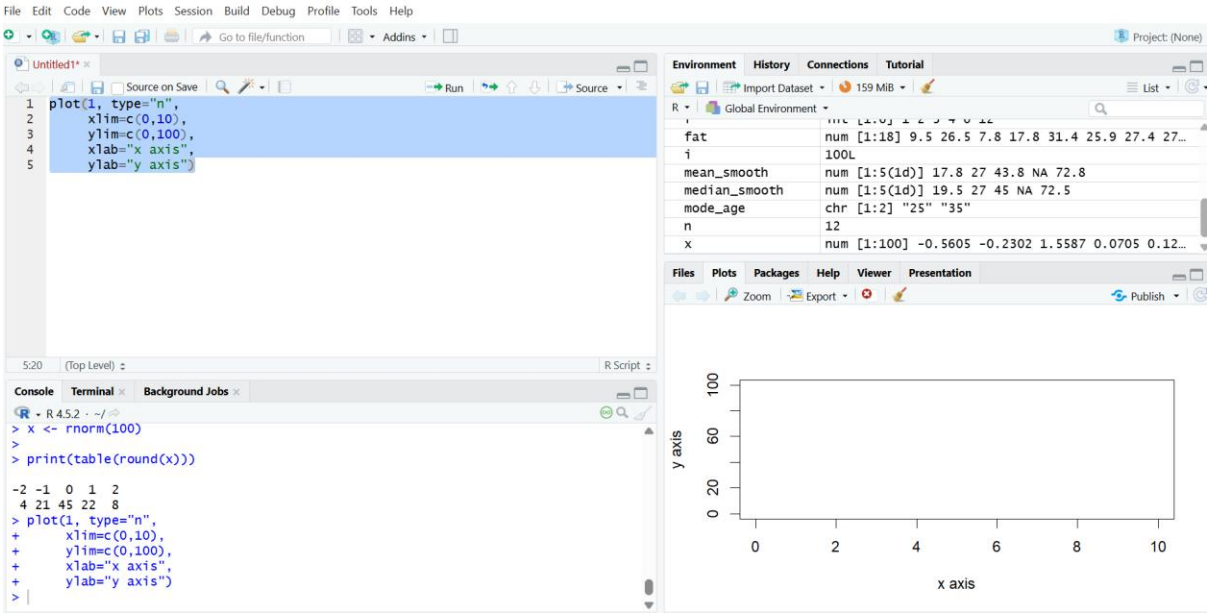
7}


print(f)
set.seed(123)
print(sample(-50:50,10,replace=TRUE))
for(i in 1:100)
{
if(i %% 15 == 0)
print("FizzBuzz")
else if(i %% 3 == 0)
print("Fizz")
else if(i %% 5 == 0)
print("Buzz")
else
print(i)
}
[1] "FizzBuzz"
[1] 91
[1] 92
[1] "Fizz"
[1] 94
[1] "Buzz"
[1] "Fizz"
[1] 97
[1] 98
[1] "Fizz"
[1] "Buzz"

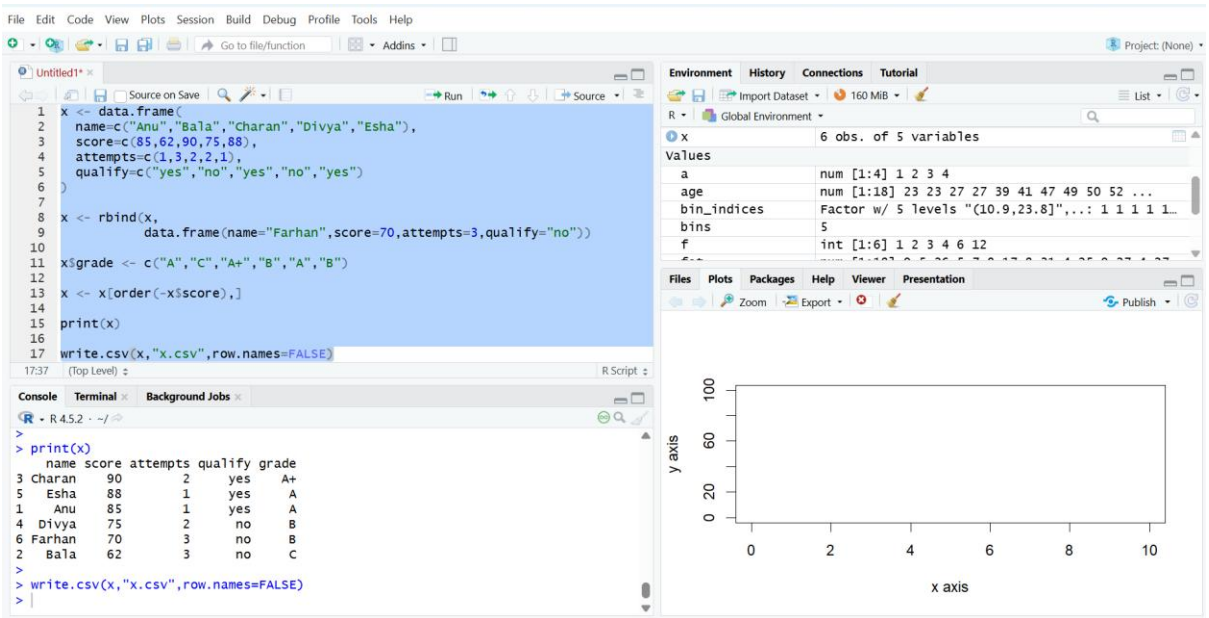
8}


set.seed(123)
x <- rnorm(100)
print(table(round(x)))
[1] 98
[1] "Fizz"
[1] "Buzz"
> set.seed(123)
> x <- rnorm(100)
> print(table(round(x)))
-2 -1 0 1 2
4 21 45 22 8

9}


File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 plot(1, type="n",
2 xlim=c(0,10),
3 ylim=c(0,100),
4 xlab="x axis",
5 ylab="y axis")
Environment History Connections Tutorial
fat num [1:18] 9.5 26.5 7.8 17.8 31.4 25.9 27.4 27…
i 100L
mean_smooth num [1:5(1d)] 17.8 27 43.8 NA 72.8
median_smooth num [1:5(1d)] 19.5 27 45 NA 72.5
mode_age chr [1:2] "25" "35"
n 12
x num [1:100] -0.5605 -0.2302 1.5587 0.0705 0.12…
Files Plots Packages Help Viewer Presentation
Console Terminal Background Jobs
> x <- rnorm(100)
> print(table(round(x)))
-2 -1 0 1 2
4 21 45 22 8
> plot(1, type="n",
+ xlim=c(0,10),
+ ylim=c(0,100),
+ xlab="x axis",
+ ylab="y axis")
y axis
x axis

10}


File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 x <- data.frame(
2 name=c("Anu","Bala","Charan","Divya","Esha"),
3 score=c(85,62,90,75,88),
4 attempts=c(1,3,2,2,1),
5 qualify=c("yes","no","yes","no","yes")
6)
7
8 x <- rbind(x,
9 data.frame(name="Farhan",score=70,attempts=3,qualify="no"))
10
11 x$grade <- c("A","C","A+","B","A","B")
12
13 x <- x[order(-x$score),]
14
15 print(x)
16
17 write.csv(x,"x.csv",row.names=FALSE)
Environment History Connections Tutorial
x 6 obs. of 5 variables
Values
a num [1:4] 1 2 3 4
age num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
bin_indices Factor w/ 5 levels "(10.9,23.8]",..: 1 1 1 1 1…
bins 5
f int [1:6] 1 2 3 4 6 12
Files Plots Packages Help Viewer Presentation
Console Terminal Background Jobs
> print(x)
name score attempts qualify grade
3 Charan 90 2 yes A+
5 Esha 88 1 yes A
1 Anu 85 1 yes A
4 Divya 75 2 no B
6 Farhan 70 3 no B
2 Bala 62 3 no C
> write.csv(x,"x.csv",row.names=FALSE)
y axis
x axis